What AcuPebble, MAM, Dual A, and PSG actually measure

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> **Ox lock:** AcuPebble **always** uses finger ox (AHI + ODI). MAM, ANR, and Dual A **do not**. Dual A SpO_2 is temple PPG. Oralable does not replace AcuPebble AHI.

Words we use

Word	Meaning
Instrument	The device and what it senses. AcuPebble: neck sound + **finger** oxygen (AHI / ODI). Oralable (MAM): temple light and motion; oxygen is **temple**, not finger. ANR: electrical muscle activity on the same temporalis. PSG-AV: the lab study.
Methods	How we write the measurement: site, sample rate, what we scored, what we do not claim. Not your MAD titration method.
Dual A	Oralable and ANR on the same anterior temporalis at the same time. Seat Oralable first. Confirm an infrared trough on a hard clench. Then add ANR, electrodes vertical. What we score today: an EMG burst, then an infrared trough about 1–5 seconds later. Not AHI. Not a night. One engineering pack is about six minutes.
Muscle versus infrared	Dual A's main pair. **Muscle** = ANR EMG (the clock). **Infrared** = Oralable IR-DC trough (blood squeezed from the tissue; it lags). Same event, two physics. Not the same units.
Paper A	The Oralable kit / feasibility paper (IEEE draft): wear, temple vitals, Dual A as a descriptive precursor. About five people at Beacon. Not a registered trial. Not an AHI claim. It does not test the FEP theory.
FEP paper	Owens and Mayoral, *Frontiers in Behavioral Neuroscience*, 2026. Theory: sleep bruxism as predictive regulation. No Oralable cohort. Hypothesis 3.5.4 needs apnea/hypopnea **times** plus a muscle clock. That coupling is not built. Cite for Paper B / Arm P, not as Paper A methods.

Instruments measure. Dual A pairs muscle with infrared. Methods is how we write that. Paper A is the kit paper. The FEP paper is the theory paper.

Why this note

You already run **AcuPebble** for home OSA. Oralable (**MAM**) sits on the **temple**. ANR is **EMG** on the same temporalis belly. Dual A is MAM + ANR on that site. These are different physics. Mixing them in methods copy makes AHI look like temple SpO_2, or EMG look like an IR trough.

Three oxygen numbers (do not merge):

Name	What it counts	Needs scored apneas/hypopneas?
AHI	Event **count** per hour of sleep	Yes
HB (Azarbarzin hypoxic burden)	SpO_2 **area tied to those events**	**Yes**
SASHB	Oralable area when temple $SpO_2 < 90\%$	**No** — temple PPG only, **not** finger ox

Your AcuPebble headline is **AHI / ODI**. That is not Azarbarzin HB unless your export has event-tied SpO_2 area. We do not claim that.

Relevance to you

Your work	What the tables say
AcuPebble (today)	Home AHI + ODI from neck + **finger** SpO_2. Keep this as the apnea reference. Oralable does not replace it.
Arm P / MAD nights	MAM can sit beside AcuPebble for temple vitals and later jaw-load. Temple SpO_2 ≠ finger ox. SASHB ≠ AHI ≠ HB.
Dual A (research)	Same-site **muscle (EMG)** then **infrared (IR-DC)** trough (~1–5 s lag). One engineering pack: IR↔EMG F1 ≈ 0.61 vs EMG; F1 vs Protocol A labels = 0; ~6 min, not a night.
FEP / hypothesis 3.5.4	Needs **event timestamps** plus a bout clock. Nightly AHI is not enough. Closest home path: AcuPebble events + Dual A. **Not built.** Cite FEP for Paper B / Arm P, not Paper A methods.
PSG-AV	Gold AHI, RMMA, and HB if scored. Lab, not the home stack we ship.

Paper A is feasibility / Dual A precursor. It does **not** test FEP endotypes.

Table 2 — what each stack gives you now

Stack	You get now	You still lack
MAM (Oralable temple)	IR trough, % drop, SASHB, tonic/phasic/rescue/recovery **labels**, Core ML, **temple** SpO_2	Finger ox; AHI; HB; electrical onset; FEP latency
ANR	EMG onset + amplitude bouts	Finger ox; AHI; HB; IR; Core ML; rescue-as-airway
Dual A	**Muscle (EMG)** versus **infrared (IR-DC)** trough + **temple** SASHB shown beside EMG	Finger ox; AHI; HB; Core ML vs EMG; overnight tonic/phasic F1; FEP latency
AcuPebble	AHI + ODI / **finger** SpO_2 (**always**)	HB unless export is event-tied; all jaw rows
AcuPebble + Dual A	AHI **plus** EMG→IR. Finger SpO_2 is **AcuPebble only**; Dual A still temple + EMG	Coupling only if event timestamps align — **not built**. Dual A SpO_2 is not the finger channel
PSG-AV	AHI **and** HB (if scored) **and** RMMA	Oralable IR-DC / Core ML / SASHB unless MAM is worn

Table 1 — constructs (what matches what)

Now = what exists today. **MAM if verified** = the name we may use only after Dual A (or PSG) F1, or an event clock. **Still not** = even after a pass.

#	Construct	Now	PSG-AV	AcuPebble	ANR	Dual A now	MAM if verified	Still not
1	**AHI**	Impossible from MAM/ANR	**Gold**	Home AHI	No	No	**None**	Temple PPG/ACC/IR cannot become AHI

#	Construct	Now	PSG-AV	AcuPebble	ANR	Dual A now	MAM if verified	Still not
2	**HB** (Azarbarzin)	Impossible (no finger ox, no scored events)	**Yes** if event-tied finger/PSG area	AHI/ODI ≠ HB unless export is event-tied — **not claimed**	No	No	**None** as HB	SASHB; Dual A pairing; AHI; ODI
3	**SASHB**	Temple PPG SpO_2<90 area — **not finger ox**	Not this formula	Not AcuPebble's formula	No (no SpO_2)	Temple SASHB shown beside EMG. Not AHI, not HB, not finger ox	Stay "temple SpO_2<90 area"	AHI; HB; ODI; finger SpO_2
4	**LP IR-DC trough**	Scored Dual A	No optical OMG	No	No	**Primary pair:** EMG first, IR later	Occlusion onset / trough (~1–5 s lag)	EMG onset; μV
5	**EMG burst onset**	ANR / PSG	**Yes** — RMMA clock	No	**Yes**	Clock for row 4	MAM cannot get electrical onset	Millisecond lockstep
6	**IR-DC % drop**	Gate ≥8% of rest	No	No	No	EMG ≥70 **and** IR ≥8%	Occlusion depth (% of rest)	EMG amplitude; %MVC
7	**EMG amplitude**	ANR 0–1023	μV / often ≥10% MVC	No	**Yes**	Contact proof, not MVC	—	Depth ≠ μV
8	**Overnight tonic**	MAM **label**	Tonic RMMA (different def)	No	Sustained EMG, not scored vs IR	Not F1'd	Tonic occlusion minutes	Tonic RMMA until F1
9	**Overnight phasic**	MAM **label**	Phasic RMMA	No	Phasic EMG	**Not F1'd** — ACC ≠ EMG	High-motion bouts; this row may fail	Phasic RMMA; grinding EMG
10	**Core ML MAM**	Product classes; 10% IR-DC **gates** only	No	No	No	Never run vs ANR	1 s optical–motion class if F1 vs EMG holds	EMG class; apnea class
11	**Rescue**	MAM state. EMG ∩ **temple** desat. **No finger ox**	Events + RMMA (finger ox)	AHI events + **finger** desat if exported	No	Pairing ≠ rescue = EMG	Load + temple-desat bout if timestamps match AcuPebble/PSG event ends	AHI; finger-ox desat; FEP airway RMMA until scored
12	**Recovery**	IR-DC + ACC settle	Could define EMG + autonomic return	No	Unused for this	Not FEP latency	Optical–motion return time	**Homeostatic latency** (your paper: EMG + HR/HRV)
13	**TFI**	Optical load index	No	No	No	No	Night occlusion burden if vs EMG episode index	MVC%; AASM SB index

Engineering Dual A pack (locked, not a night): session `20260812_085110`, ~6 min. Median EMG→IR lag ≈ **4.9 s**. IR↔EMG F1 ≈ **0.61**. F1 vs Protocol A labels = **0**.

Do / do not (for methods)

Do	Do not
Keep AcuPebble for AHI / ODI (finger ox)	Call Oralable or Dual A an HSAT or finger-ox channel
Call Dual A primary pair EMG → IR trough (muscle versus infrared)	Call the IR trough EMG; call ACC phasic RMMA
Call tonic / phasic / rescue / Core ML **labels** until F1	Treat Core ML as scored vs ANR
Call recovery optical–motion settle	Call it homeostatic latency
Call AcuPebble AHI/ODI	Call AHI or ODI Azarbarzin HB unless the export has event-tied SpO_2 area

Device-inferred wellness / research signals — not a medical diagnosis. Paper A does not test FEP. Canonical lock: MEASUREMENT_CONSTRUCT_MAP (data room). Do not paste Table 1 into Paper A methods — point here.